



Exercises session 3

Polynomial time complexity II

1 Definition

Definition 1.1 (Vertex colouring) A vertex colouring of a graph $G = (V, E)$ with k colours is a function

$$c : V(G) \rightarrow \{1, \dots, k\}$$

such that adjacent nodes have different colours, that is

$$(u, v) \in E \rightarrow c(u) \neq c(v)$$

Definition 1.2 (Conjunctive normal form) A propositional logic formula φ is in conjunctive normal form (CNF) if

$$\varphi = C_1 \wedge \dots \wedge C_m$$

where each C_i is a disjunction of literals

$$C_i = (\ell_{i1} \vee \dots \vee \ell_{ik})$$

and a literal is either a variable or the negation of a variable.

We also say that a formula is in k -CNF if it is in CNF and each clause has at most k literals.

2 Session exercises

1. Consider the following formula:

$$\varphi = (\neg x_1 \vee x_2) \wedge (\neg x_2 \vee x_3) \wedge (\neg x_3 \vee x_4) \wedge (\neg x_4 \vee x_1) \wedge (x_1 \vee x_5) \wedge (\neg x_5)$$

- (a) Determine if the formula is satisfiable using the approach seen for the 2-SAT problem
- (b) Repeat previous step, but now using the approach seen for HORN-SAT
- (c) Compare the running times of both methods.

2. Consider the following problem:

$$2\text{-COL} = \{G : G \text{ has a 2-colouring}\}$$

- Design an algorithm that decides if a graph has a 2-colouring
 - Relate this problem to 2-SAT and conclude that 2-COL is in PTIME.
3. We saw in classes that Horn-SAT is solvable in polynomial time. However, sometimes not all clauses can be satisfied. In such cases, we seek an assignment that satisfies the maximum number of clauses—this is the **Max-Horn-SAT** problem.
- Given the following Horn formula:

$$\varphi = (\neg x_1 \vee x_2) \wedge (\neg x_2 \vee x_3) \wedge (\neg x_3 \vee x_4) \wedge (\neg x_4 \vee x_1) \wedge (\neg x_3) \wedge (x_4)$$

- Determine if a satisfying assignment exists.
- If the formula is unsatisfiable, determine the **maximum number of clauses** that can be satisfied.
- Provide an example of an assignment that **maximizes satisfied clauses**.
- Can you think of an efficient algorithm for solving Max-Horn-SAT?